

Answers and worked solutions

Quick answer key

1	A	11	B	21	E	31	A
2	B	12	A	22	A	32	B
3	C	13	D	23	C	33	E
4	C	14	A	24	E	34	E
5	D	15	E	25	C	35	D
6	E	16	A	26	A	36	D
7	B	17	D	27	B	37	D
8	A	18	B	28	B	38	C
9	E	19	E	29	C	39	C
10	B	20	C	30	D	40	D

Timing and difficulty

The estimated solution times total about 57 minutes. They are editorial estimates for prepared candidates, not measured student timings. Trial the questions before using them to predict scores.

These 40 items target missing content; their topic proportions should not be treated as official ESAT weightings.

Each question has one correct answer. The answer letters are balanced across A-E.

Coverage and specification map

Topic	Count	Questions
Electromagnetic induction	5	1, 11, 19, 26, 32
Circuit components	5	2, 12, 20, 27, 40
Springs and elasticity	4	3, 15, 25, 36
Radioactivity	4	4, 14, 29, 37
Transformers and transmission	4	5, 13, 23, 33
Electrostatics	3	6, 17, 34
Refraction	4	7, 18, 28, 38
Particle models and states	3	8, 22, 31
Basic magnetism	3	9, 21, 30
Sound and ultrasound	3	10, 24, 39
Electromagnetic spectrum	2	16, 35

Design notes

Short stems, five choices per item, no calculator arithmetic and 17 grayscale diagrams. The set mixes graph reading, experimental evidence, two-stage calculations and qualitative reasoning.

Scope: ESAT Physics with Mathematics 1. No differentiation, Snell's law, quantitative charged-particle motion, internal resistance, resistivity, friction coefficients or capacitor theory is needed.

The editable source records each question's specification references, suggested time, answer and worked solution.

1 | Answer A

Electromagnetic induction · P2.4(a-c) · suggested 75 s

Changing the approaching pole reverses the voltage, so the first stage is negative. A stationary magnet gives no changing field and hence zero voltage. Reversing its motion reverses the voltage again, giving positive.

2 | Answer B

Circuit components · P1.2(f-g,i,m) · suggested 95 s

At $I = 0.30$ A, the graph gives $V_{\text{lamp}} = 4.0$ V. The resistor voltage is $IR = 3.0$ V, so this operating point uses the full 7.0 V. Thus $P_{\text{lamp}} = VI = 4.0 \times 0.30 = 1.2$ W. The total supply power is not the lamp power.

3 | Answer C

Springs and elasticity · P3.3(c-d) · suggested 90 s

The spring constant is $k = 6.0/0.030 = 200$ N m⁻¹. Subtract the initial stored energy from the final stored energy:

$$\Delta E = \frac{1}{2}(200)(0.050^2 - 0.020^2) = 0.21 \text{ J.}$$

Using only the change in extension in $\frac{1}{2}kx^2$ would be incorrect.

4 | Answer C

Radioactivity · P7.3(a,d) · suggested 80 s

Paper causes no reduction, so no detected alpha component is present. Aluminium removes a beta component but leaves gamma plus background. The gamma contribution is $90 - 20 = 70$ counts min⁻¹.

5 | Answer D

Transformers and transmission · P2.5(b-c); P1.2(m) · suggested 90 s

The secondary voltage is $240 \times 40/800 = 12$ V, as required. The maximum input power is $240 \times 0.15 = 36$ W. For an ideal transformer the output power is also 36 W, enough for $36/6 = 6$ lamps.

6 | Answer E

Electrostatics · P1.1(b-d) · suggested 65 s

The negatively charged droplets repel one another. They deliver excess electrons to the panel, reducing its positive charge. Electrons must therefore leave through the connection to maintain the panel's positive charge.

7 | Answer B

Refraction · P6.2(b-c); P6.3(c-e) · suggested 75 s

The angles to the normal are $90^\circ - 35^\circ = 55^\circ$ in X and $90^\circ - 60^\circ = 30^\circ$ in Y. The ray bends towards the normal, so its speed decreases. Its frequency is unchanged at the boundary.

8 | Answer A

Particle models and states · P5.1; P5.3 · suggested 80 s

The constant temperature means that average particle kinetic energy is unchanged. Energy is still supplied and is used to change the arrangement and separation of particles during melting, increasing their potential energy.

9 | Answer E

Basic magnetism · P2.2(a-c) · suggested 80 s

The field around P is anticlockwise, so at M, to its right, the field points upwards. Reversing Q makes its field clockwise; at M, to its left, that field also points upwards. The two equal fields add to $2B$.

10 | Answer B

Sound and ultrasound · P6.1(g); P6.4(g) · suggested 85 s

The extra return time is $70 - 30 = 40 \mu\text{s}$. It corresponds to travelling through the layer twice. Hence

$$d = \frac{1500 \times 40 \times 10^{-6}}{2} = 0.030 \text{ m} = 3.0 \text{ cm}.$$

11 | Answer B

Electromagnetic induction · P2.4(d-e); P6.1(f) · suggested 90 s

The original period is 8.0 ms , so the first positive maximum is at 2.0 ms . Doubling the rotation speed halves all corresponding times, giving 1.0 ms . Faster cutting of the magnetic field produces a larger peak voltage.

12 | Answer A

Circuit components · P1.2(h-i,f) · suggested 100 s

The fixed resistor has 8.0 V , leaving 4.0 V across the LDR. Their current is the same, so $R_{\text{LDR}}/3.0 = 4.0/8.0$, giving $1.5 \text{ k}\Omega$. More light reduces the LDR resistance, increasing current and the fixed-resistor voltage.

13 | Answer D

Transformers and transmission · P2.5(d); P1.2(m) · suggested 90 s

Initially $I = 20000/2000 = 10$ A, so the loss is $I^2 R = 500$ W. Afterwards $I = 20000/10000 = 2$ A, so the loss is 20 W. The reduction is $500 - 20 = 480$ W.

14 | Answer A

Radioactivity · P7.2(d); P7.3(c) · suggested 75 s

Alpha particles are positive and are attracted towards the lower negative plate. Beta-minus particles are negative and are attracted towards the upper positive plate. Uncharged gamma radiation is not deflected.

15 | Answer E

Springs and elasticity · P3.3(c-d); P3.2(c-d) · suggested 100 s

Let each spring constant be k . In X, each spring carries W , so the total energy is $2[W^2/(2k)] = W^2/k$. In Y, each carries $W/2$, so the total energy is $2[(W/2)^2/(2k)] = W^2/(4k)$. The ratio is 4 : 1.

16 | Answer A

Electromagnetic spectrum · P6.5(b-e) · suggested 65 s

The thermal camera detects infrared. The frequency order is infrared, ultraviolet, X-rays. All three are electromagnetic waves and travel at the same speed in a vacuum, despite their different frequencies.

17 | Answer D

Electrostatics · P1.1(b); P1.2(d) · suggested 80 s

Overall the object has lost 3.0×10^{10} electrons, so it is positive. The charge is $(3.0 \times 10^{10})(1.6 \times 10^{-19}) = 4.8 \times 10^{-9}$ C = +4.8 nC.

18 | Answer B

Refraction · P6.1(h); P6.2(b-d) · suggested 85 s

Five crests span four wavelengths: $\lambda_d = 24/4 = 6$ cm. Thus $f = 30/6 = 5$ Hz. Frequency is unchanged by refraction, and $\lambda_s = 16/4 = 4$ cm, so $v_s = 5 \times 4 = 20$ cm s⁻¹.

19 | Answer E

Electromagnetic induction · P2.4(a-c) · suggested 85 s

Only a changing current in P produces a changing magnetic field and an induced voltage in Q. The current increases in I and decreases in III, so the induced voltages are non-zero with opposite signs. The constant currents in II and IV give zero voltage, even though the current in II is non-zero.

20 | Answer C

Circuit components · P1.2(f,h-i,m) · suggested 90 s

Initially $I = 12/6000 = 2.0 \text{ mA}$, giving $P_T = I^2 R_T = 16 \text{ mW}$. Afterwards $I = 12/3000 = 4.0 \text{ mA}$, giving $P_T = (0.004)^2(1000) = 16 \text{ mW}$. The current doubles while the thermistor resistance falls by four.

21 | Answer E

Basic magnetism · P2.1(a,c-d) · suggested 80 s

The nearby north pole induces a south pole at X and a north pole at Y. Outside a north pole the field points away, so the compass north end points away from Y. Soft iron is useful because its induced magnetisation largely disappears when the applied field is removed.

22 | Answer A

Particle models and states · P5.1; P5.3(a) · suggested 70 s

Particles have a range of kinetic energies. Those with enough energy can escape from the surface. Preferential loss of these higher-energy particles reduces the average kinetic energy, and hence the temperature, of the remaining water.

23 | Answer C

Transformers and transmission · P2.5(b-c); P1.2(m) · suggested 105 s

The secondary voltages are 12 V and 24 V. The load powers are $12^2/6 = 24 \text{ W}$ and $24^2/24 = 24 \text{ W}$. The input supplies both: $I_p = (24 + 24)/120 = 0.40 \text{ A}$.

24 | Answer E

Sound and ultrasound · P6.1(e-f); P6.4(c) · suggested 85 s

For P, $T_P = 4 \times 0.50 = 2.0 \text{ ms}$. For Q, $T_Q = 2 \times 0.50 = 1.0 \text{ ms}$, so its frequency is doubled. Both voltage amplitudes are $3 \times 0.20 = 2 \times 0.30 = 0.60 \text{ V}$; hence the sound amplitudes are equal.

25 | Answer C

Springs and elasticity · P3.3(a-b) · suggested 95 s

At 2.0 N, the graph gives an extension of 3.0 cm relative to the original length, so the length is 13.0 cm. When the force reaches zero, 2.0 cm of extension remains permanently.

26 | Answer A

Electromagnetic induction · P2.4(a-c) · suggested 80 s

In motion 1 the rod cuts field lines so that opposite charges are separated towards its two ends. Moving along its own length does not produce end-to-end charge separation. Moving parallel to the field does not cut the field lines. Only motion 1 gives an end-to-end induced voltage.

27 | Answer B

Circuit components · P1.2(h-k,m) · suggested 105 s

Reversal blocks the diode branch completely. The remaining conducting path contains $4.0\ \Omega$ and $12\ \Omega$ in series, so $R = 16\ \Omega$. Hence $I = 12/16 = 0.75\ \text{A}$ and $P = VI = 9.0\ \text{W}$. The blocked branch is an open circuit, not a short circuit.

28 | Answer B

Refraction · P6.1(g); P6.2(b-c) · suggested 100 s

The original transit time is $0.12/(2.0 \times 10^8) = 0.60\ \text{ns}$. The replacement gives $0.12/(1.5 \times 10^8) = 0.80\ \text{ns}$. Subtracting gives an extra delay of $0.20\ \text{ns}$. No bending occurs at normal incidence, but speed still changes.

29 | Answer C

Radioactivity · P7.3(a,e); P7.4(b) · suggested 90 s

Beta radiation is sensitive to the relevant thickness changes. A long half-life avoids rapid changes in source output. A thinner sheet absorbs fewer beta particles, so the detected count rate increases.

30 | Answer D

Basic magnetism · P2.1(b); P2.2(b) · suggested 85 s

Anticlockwise current makes the viewed left end a north pole, so the right end is a south pole. Just to the right of a south pole, the field points left towards it. Reversing the current reverses the poles and the field, so the compass then points right.

31 | Answer A

Particle models and states · P5.1(b); P5.2(a-b) · suggested 75 s

Heating increases average particle kinetic energy and speed. More frequent and more forceful wall collisions increase momentum transferred per unit time, hence force and pressure. The fixed particle number and container volume rule out an increase in particle number or average spacing.

32 | Answer B

Electromagnetic induction · P2.4(e-f); P1.2(d,f,h); M4.14 · suggested 100 s

The peak current in the positive half-cycle is $4.0/2000 = 2.0 \text{ mA}$. Its current-time graph is a triangle of base 4.0 ms . The charge is its area:

$$Q = \frac{1}{2}(4.0 \times 10^{-3})(2.0 \times 10^{-3}) = 4.0 \mu\text{C}.$$

The diode blocks the negative half-cycle, so it does not cancel this charge.

33 | Answer E

Transformers and transmission · P2.5(b-c); P1.2(m) · suggested 100 s

Halving the primary turns doubles the secondary voltage. With the same load resistance, secondary current also doubles, so output power quadruples. Input power therefore quadruples; since primary voltage is fixed, primary current increases by a factor of four.

34 | Answer E

Electrostatics · P1.1(b-d) · suggested 80 s

Positive charging means a loss of electrons, not a gain of protons. Earthing allows electrons to enter from Earth and replace that deficit. The conducting path prevents a large charge build-up that could cause a spark.

35 | Answer D

Electromagnetic spectrum · P6.1(g); P6.5(a,e) · suggested 80 s

Both microwave and infrared signals travel at the same speed in vacuum. Add the two path lengths, rather than their difference:

$$t = \frac{(36000 + 45000) \times 1000}{3.0 \times 10^8} = 0.27 \text{ s}.$$

36 | Answer D

Springs and elasticity · P3.3(d); P3.7(c,f) · suggested 95 s

The initial spring energy is $\frac{1}{2}(800)(0.050)^2 = 1.0 \text{ J}$. After the rough section, 0.80 J remains. At the highest point this is gravitational potential energy:

$$h = \frac{0.80}{0.050 \times 10} = 1.6 \text{ m}.$$

37 | Answer D

Radioactivity · P7.2(b); P7.4(b) · suggested 80 s

Half-life describes the probability of decay, not an exact schedule for a small sample. Each surviving nucleus still has a one-half probability of surviving the next half-life. The expected number remaining is $72/2 = 36$, with random variation around that value.

38 | Answer C

Refraction · P6.2(b-c); P6.3(c-e) · suggested 90 s

The ray must bend towards the normal on entering glass and away from the normal on leaving. For parallel faces with air on both sides, the emergent ray is parallel to the incident ray. Only R satisfies all these conditions.

39 | Answer C

Sound and ultrasound · P6.1(a-c); P6.4(b,d) · suggested 65 s

Sound in air is longitudinal: particles move back and forth parallel to the propagation direction. Energy moves along the tube without the bulk transport of air with the wave.

40 | Answer D

Circuit components · P1.2(e-f,h-i) · suggested 95 s

The observed total resistance is $9.0/0.0045 = 2000 \Omega$, exactly the working LDR resistance. The LDR has the whole supply voltage, so the fixed resistor must have zero voltage across it while carrying current: it is short-circuited. An open LDR could give the voltage reading but would give zero current.

Source checks and editorial basis

The following official materials were consulted for content scope, pacing and question presentation. The questions in this pack are newly written; they do not reproduce questions from these papers.

ESAT format: 27 multiple-choice questions in 40 minutes per module; no calculator. The 60-minute suggestion for this 40-question supplement approximates that pace.

ESAT content specification

https://uat-wp.s3.eu-west-2.amazonaws.com/wp-content/uploads/2024/05/03165424/ESAT_Content_Specification.pdf

Official ESAT Physics Guide

https://uat-wp.s3.eu-west-2.amazonaws.com/wp-content/uploads/2026/06/30103558/ESAT_Guide_Physics.pdf

NSAA 2023 Section 1: Physics, questions 21-40

https://uat-wp.s3.eu-west-2.amazonaws.com/wp-content/uploads/2024/05/07120523/NSAA_2023_S1_Question_Paper.pdf

NSAA 2022 Section 1: Physics, questions 21-40

https://uat-wp.s3.eu-west-2.amazonaws.com/wp-content/uploads/2024/05/07120525/NSAA_2022_S1_Question_Paper.pdf

Official ESAT test format

<https://esat-tmua.ac.uk/about-the-tests/esat-test/>

All question-specific methods are shown in the worked solutions. Experimental assumptions are supplied where they affect the answer. Difficulty and timing still require student calibration.